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Contextuality of Syntax and Adieu to the A/A'-Distinction

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Abstract: The paper highlights a broader theoretical move toward contextuality in syntax, which has largely gone unnoticed because the contextuality is manifested quite differently in different syntactic domains, making the contextuality connection across them not easy to discern. In particular, the move toward contextuality is shown to be manifested in structure building, labeling, islandhood, phases, phasal edges, spell-out domains, and the EPP. The contextuality of syntax is also shown to be non-Markovian in nature, hence argues against treating syntax as Markovian in nature. One Markovian property is, however, maintained: no look-ahead. Most importantly, it is shown that the contextual approach to the EPP opens the door for eliminating the A/A'-distinction from binding, in a way where all that is needed for binding is c-command: with traditional A'-movement, the nominal simply fails to c-command the potential bindee.

Keywords: A/A'-distinction, binding, the EPP, islands, labeling, phases, spell-out domains

1. Introduction

This paper highlights a broader theoretical move toward contextuality/context-sensitivity in syntax, which has gone unnoticed for two reasons: (i) it spans different domains, hence it is not easy to notice if one's attention is focused on a single domain; (ii) the contextuality is manifested quite differently in different domains, which makes the contextuality connection across different domains difficult to discern. In particular, contextuality is shown to characterize recent developments regarding locality/islandhood, structure building, labeling, the EPP, and spell-out.

The paper has two goals beyond highlighting the contextuality of syntax. There are conflicting claims regarding whether syntax is Markovian in nature: Chomsky (2021) endorses this position, while one of the main points of Chomsky (1957) was that syntax is non-Markovian. I address the issue from a new perspective, by examining what the contextuality of syntax means for this debate.

Most importantly, I explore consequences of the contextual approach to the EPP for binding. I show that this approach naturally leads to the conclusion that elements undergoing A'-movement are actually not nominal. The paper takes advantage of this idea to develop an approach to binding that dispenses with the A/A'-distinction, where all that is needed for binding is c-command.

The paper is organized as follows. Section 2 discusses the history of research on locality and structure building, showing that it has consistently emphasized the role of contextuality. Section 3 discusses the contextuality of spell-out domains. Section 4 turns to the EPP and the more general issue of whether syntax is Markovian in nature. Section 5 shows that the contextual approach to the EPP opens the door for eliminating the A/A'-distinction from binding.

2. Locality/islandhood and structure building

In the bounding approach to locality/islandhood (Chomsky 1973), the trouble-makers for movement were defined rigidly: NP and IP were bounding nodes regardless of their syntactic context. *Barriers* (Chomsky 1986) were quite different but the importance of one difference went largely unnoticed—its contextuality. One cannot even ask whether, for example, CP in general is a barrier. This depends on its syntactic context; *Barriers* define trouble-makers for movement contextually. Chomsky's (2000) *phases* went back to the bounding approach in defining the trouble-makers (phases) rigidly, CP and vP being phases regardless of their structural position. This was soon followed by various contextual approaches, where whether XP is a phase depends on its syntactic context (on a par with *Barriers*, in contrast to the bounding/early phase approach), see Bošković (2014) and references therein. Thus, in Bošković (2013, 2014), there is a phasal domain, with the highest phrase in the domain functioning as a phase. Thus, DP is a phase in the nominal domain when present; when not, a lower projection is a phase.¹

Bošković (2016b) argues the same contextuality holds for *phasal edges*, the PIC status of a Spec also being affected by context: the highest phrase in a phasal domain is a phase, and the highest edge in multiple-edge contexts (α in (1)) functions as the phasal edge.

(1) $[XP \ \alpha \ [XP \ \beta \dots XP = \text{phase}]$

There has thus been a consistent move toward contextuality in the locality of movement. The scope of contextuality is even broader. In Chomsky (2013) labeling is also contextual: X behaves differently for labeling depending on the context. A phrase behaves differently in phrase-phrase and head-phrase mergers.

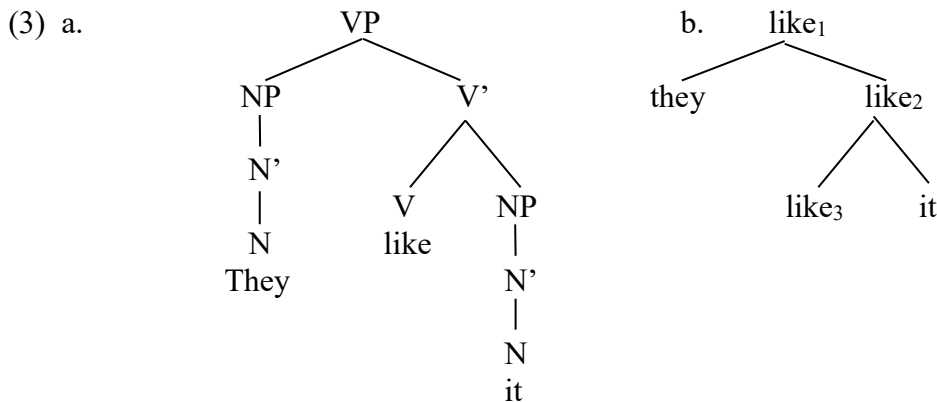
¹Its structural sensitivity enables this approach to capture cross-linguistic variation regarding extraction without parameterizing phasehood, which would involve parameterizing the computational system and which is needed under rigid phasehood, see Bošković 2014.

It also behaves differently in different phrase-phrase contexts and its labeling status changes derivationally: α in (2) changes from an unlabeled object (before *what* moves) to a CP (after the movement).

(2) What do you think [_{α} ~~what~~[?] Mary bought ~~what~~]?
 (The ~~what~~ is α before movement, the [?] is α after movement.)

Bare-Phrase structure is also contextual: whether β is a head, phrase, or an intermediate projection depends on the context. Its status also changes derivationally: what's a maximal projection after a head and a phrase merge becomes an intermediate projection with another merger.

Note the parallelism with locality: with bounding/rigid phasehood, one can look at a node, without looking at its context, and determine if it's a bounding node/phase. This is impossible in *Barriers*/contextual phasehood. In GB-structure, one only needs to look at a node to determine if it's a phrase or a head (3a). In Bare-Phrase structure, looking at any *like* in (3b) doesn't help in this respect; this is determined contextually: what doesn't project (*like*₁) is a phrase, what's not a projection (*like*₃) is a head.



The A/A'-distinction is also now contextual. To illustrate, movement from vP must proceed via Spec, vP. Its A/A'-status depends on the nature of that movement: if it is A-movement (the position below/above Spec, vP in the chain is an A-position), Spec, vP is an A-position (also if landing site of object shift); if it is A'-movement, as in *How_i/Who_i do you [_{vP} *t_i* think [*Mary left t_i*]]*, Spec, vP is an A'-position; we need to look at the larger context to determine the A/A'-status of a Spec, vP.²

Additionally, Bošković (2015, 2016a, 2018, 2020) provides a uniform account of all island/locality-of-movement effects based on a contextual approach to phases and the labeling theory, where there are no islands as this notion has been traditionally understood—there are no phrases that by their nature, independently of their syntactic context, disallow extraction, extraction being possible from all islands in well-defined contexts.

3. Spell-out

Evidence from many PF phenomena shows that what is sent to spell-out is phases, not phasal complements (Bošković 2016a), as expected since only phases have a theoretical status. However, this raises an issue for successive-cyclic movement (SCM). If what is sent to spell-out is phases and SCM targets phasal edges, as standardly assumed, X undergoing movement would get frozen since it would be part of what's spelled-out hence inaccessible. This section, based on work in progress, shows the conundrum can be resolved under the labeling theory: spell-out can correspond to phases and SCM can still target phasal edges. Crucially, spell-out is contextual in that a phase with a Spec does not always behave the same way for spell-out.

Under standard assumptions, CP is a phase in (4), and IP the spell-out domain. Under phasal complement spell-out, Y can interact with X but not Z in PF; under phasal spell-out, Y can interact with Z but not X.

²Interestingly, in **Cy_i is said [_{CP} *t_i* I like *t_i*]* *t'* is then in an A-position, which means there's no Improper/A-A'-A movement here.

(4) X [_{CP} Y [_{IP} Z

Bošković (2016a) provides a number of arguments that Y can interact with Z but not X. Consider Bulgarian/Macedonian cliticization:

- (5) a. Vera *mi go* dade. Bg/Mac:OK
 Vera me.dat it.acc gave
 ‘Vera gave it to me.’
 b. *Mi go* dade. Bg:*/Mac:OK
 c. Dade *mi go*. Bg:OK/Mac:*

Bulgarian/Macedonian clitics are verbal clitics. Bulgarian clitics encliticize, Macedonian ones procliticize. Bošković (2001) gives a copy-and-delete account of (5) based on Franks’ (1998) proposal that a lower copy can be pronounced iff necessary to avoid a PF violation. Clitics undergo movement. In Macedonian, the higher copy can always be pronounced, hence must be. In Bulgarian, it can be pronounced when something precedes it, not when nothing precedes it (6b). Here, a lower copy is pronounced to avoid a PF violation.

- (6) a. X cl V ~~cl~~ b. ~~cl~~ V cl (Bg)
 c. (X) cl V ~~cl~~ (Mac)

Conjunction can support Bulgarian enclitics, hence the impossibility of V-Cl below.

- (7) a. I *ti go* dade. b. *I dade *ti go*.
 and you.dat it.acc gave

Consider *wh*-C/*li*-questions. They involve V-to-C movement, V carrying the clitics along, as in Macedonian (8). Surprisingly, in Bulgarian, the conjunction cannot support the enclitics. Instead of expected (9b), we get (9a). Why does lower-copy pronunciation apply in (9a), in contrast to (7a)?

- (8) I *ti go* dade *li*?
 and you.dat it.acc gave Q
 ‘And did (s)he give it to you?’
 (9) a. I dade *li ti go*? b. *I *ti go* dade *li*?
 and gave Q you.dat it.acc

Franks and Bošković (2001) provide a multiple-spell-out account, where what is spelled out is crucially the phase. The Cl+V-cluster head-moves to *li*. CP is then sent to spell-out, with lower enclitic copies pronounced since there is nothing in (10) that can support them.

(10) [_{CP} [_C ~~ti go~~ dade+*li*] *ti go dade*]

The conjunction is only then added, deriving (9a)/(11):

(11) i [_{CP} [_C ~~ti go~~ dade+*li*] *ti go dade*]

If phasal complements were sent to spell-out, C would be in the same spell-out domain as the conjunction. This should lead to higher copy pronunciation even in Bulgarian. This does not happen (see (9b)), indicating the CP phase is spelled out.

Turning to Abruzzese raddoppiamento fonosintattico (RF), where a word’s initial consonant undergoes gemination conditioned by the preceding word, D’Alessandro and Scheer (2015) show RF is spell-out domain sensitive based on active/passive contrasts. Importantly, in (12), ‘whom’ in Spec, CP triggers RF within IP, which shows phasal complements are not spell-out domains (RF is not possible across CP).

- (12) lu wagliōnə chə ttu si vistə
 the boy whom you are seen

PF interaction is thus impossible between the edge of phase XP and material outside XP but possible between the XP-edge and the complement of X (see Bošković 2016a for additional arguments). The conclusion is then that spell-out domains correspond to phases. What about successive-cyclic-movement (SCM)? If phases are spell-out units and SCM targets phasal edges, X undergoing SCM would get frozen: it would be part of a spelled-out unit, hence inaccessible. The labeling theory, however, resolves the issue.

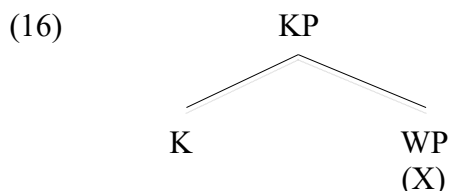
Chomsky (2013) argues that when a head and a phrase merge, the head labels the resulting object. When phrases merge, there are two ways to label: feature-sharing or movement, traces being ignored for labeling. (13) illustrates the former: when *what* merges with CP, both have the Q-feature, which determines the resulting label.

- (13) I wonder [_{CP} what_i [_C C [John bought t_i]]]

Consider SCM (14). SCM doesn't involve feature-sharing (which follows Bošković 1997, 2007, 2008). Since there is no feature-sharing between *that* and *what*, the embedded clause cannot be labeled when *what* moves to its edge (15). After *v* merges, *what* moves. The element merged with *that*-CP being a trace, it's ignored for labeling, with ? labeled as CP.

- (14) What_i do you think [_{CP} t'_i [_C that [John bought t_i]]]?
 (15) v think [? what [_{CP} that [John bought t_i]]]

Recall Bošković's (2014) contextual approach to phases, where there are phasal domains with the highest phrase in the domain a phase. *Phase* is determined when the domain is closed. This is in turn determined when the element with the relevant phasal property (X) no longer projects, which closes the phasal domain—X determines a phase when X no longer projects. WP in (16) then becomes a phase only after merging with K, with K projecting (i.e. with WP no longer projecting).



So, the phasal status of WP is determined only after merging with K.

Consider again SCM:

- (17) What do you think that John bought?
 (18) [? what [_{CP} that [_{IP} John bought]]]

That and IP merge, with the resulting object labeled CP (18). *What* then merges with it. The resulting object is unlabeled. Since there was a further merger and CP did not project, the phasal domain is closed. As the highest projection in the phasal domain, the CP sister of *what* is a phase and sent to spell-out, given that what is spelled out is phases. (Phases/spell-out domains are bolded in (18)/(20)). Once *what* moves away (after additional structure merges), traces being ignored for labeling, ? is labeled as CP.

We can then have spell-out correspond to phases and SCM target phases. *What* targets CP (*that*+IP), and this CP is sent to spell-out. *What* is outside the spell-out domain, but not because of the PIC, which is not needed, but because it is not part of the phase that is spelled out.

Consider agreeing Specs:

- (19) Mary wonders what John bought.
(20) [_{CP} what [_{CP} +*wh*-C [_{TP} John bought]]

There is feature-sharing when *what* and CP merge. The resulting object is determined to be a phase after merger with *wonder*, and sent to spell-out.

Traditionally, in the embedded clause *what* in (13) and (14) is in the same position, and at the phase edge (outside the spell-out domain), but in the above analysis they differ regarding whether *what* is in the spell-out domain. What is particularly important is that phase XP with YP in its Spec does not always behave the same way for spell-out, indicating contextuality of spell-out: with SCM, the sister of YP is sent to spell-out, otherwise the full XP is, although both involve phasal spell-out.

4. The EPP

This section discusses the EPP. What is relevant is the approach from Bošković (2024), where there is an EPP domain, with the EPP satisfied in the highest projection in the domain. This approach shows that contextuality is also relevant for the EPP, actually in the same way as for phases and phasal edges. The approach is based on a subject-*wh*-movement puzzle.

4.1. Subject *wh*-movement

It is debated whether *who* in (21) is in Spec, CP, moving via Spec, IP, or in Spec, IP.

- (21) Who left?

There is evidence against both positions. Consider evidence for *wh*-movement (see also Bošković 2024, Messick 2020).³ (22) shows *hell*-phrases modify only *wh*-moved DPs. (23) shows they can modify subject *wh*-phrases (Pesetsky 1987), indicating they undergo *wh*-movement.

- (22) a. *Who left who the hell? b. Who the hell did he leave?
(23) Who the hell left?

Inverse scope in (24a) shows object *everyone* can scope over a quantifier in Spec, IP. This is impossible in (24b), which indicates *who* is not in Spec, IP in (24b) (Mizuguchi 2014).

- (24) a. Someone loves everyone. (someone>everyone/everyone>someone)
b. Who loves everyone? (who>everyone/*everyone>who)

Turning to evidence against *wh*-movement via Spec, IP, in many languages with both SV and VS orders, where S doesn't move to Spec, IP in VS, the orders have different morphology. Importantly, (21) has the VS morphology (e.g. Fiorentino). This shows subject *wh*-movement doesn't proceed via Spec, IP or we would get the SV morphology.

West Ulster English provides particularly strong evidence. It allows Q-float with *wh*-movement (*What_i did he say all t_i that he wanted?*). It is also possible in (25a). However, (25b) is disallowed. (25b) shows a subject in Spec, IP cannot float *all* post-verbally in passives. This rules out the derivation where *who* in (25a) moves to Spec, CP via Spec, IP (or stays in Spec, IP) since *all* then floats under movement to Spec, IP, which is disallowed (25b).

- (25) a. Who was arrested all in Duke Street?
b. *They were arrested all last night. (McCloskey 2000)

³Regarding *do*-support, it is a last-resort operation to support a T-affix when a non-null element intervenes between it and V. There is no such intervener in (21), hence no *do*-support.

So, *who* doesn't pass through Spec, IP in (21). How is the EPP satisfied then? Bošković (2024) addresses the issue by arguing there are two *wh*-positions, with the lower one occupied by *wh*-moved subjects.

4.2. Two *wh*-positions⁴

Kaisse (1983) shows there is a one-word-host restriction on aux-contraction hosted by moved *wh*-phrases (26)-(27), but crucially, not with subject *wh*-phrases (28). This shows the *wh*-phrases/auxiliaries are not in the same position in (27) and (28). (Recall the subject *wh*-phrase does *wh*-move.)

- (26) a. What's Mary buying? b. When's dinner?
(27) *Which man's she the fondest of?
(28) Which man's leaving first?

Further, (29) shows only the landing site of non-subject *wh*-movement is higher than topics, which means *wh*-moved subjects are lower than *wh*-moved objects.

- (29) a. ?Mary wonders which book, for Kim, Peter should buy.
 b. *Mary wonders which student, for Kim, should buy that book.

Polarity adverbs also indicate different landing sites for non-subject and subject *wh*-movement:

- (30) a. *What under no circumstances should Mary ever buy?
 b. Who under no circumstances should ever hire Peter?
(31) a. *What should under no circumstances Mary ever buy?
 b. Who should under no circumstances ever hire Peter?

Importantly, long-distance-moved subjects pattern with objects: aux-contraction in (32) patterns with (27), not (28), which means only *local* subjects *wh*-move to the lower position.

- (32) *Which man's Peter claiming will leave first?

The same obtains with topicalization (compare (33) and (29b)) and polarity adverbs ((34)-(35) pattern with (30a)/(31a) not (30b)/(31b)).

- (33) ?I wonder which student, for Kim, Mary said should buy that book.
(34) *Who under no circumstances should Ann ever say stole it?
(35) *What should under no circumstances Mary ever believe destroyed it?

In many languages, C is different in local subject and other questions.⁵ In Norwegian, the relevant head in subject *wh*-questions is *som*, but it's null in object/adjunct/long-distance-subject questions. The obvious conclusion is that the *wh*-phrases in (36) and (37) don't land in the Spec of the same head.

- (36) Vi vet hvem *(som) snakker med Marit.
 we know who C talks with Marit
(37) a. Vi vet hvem (*som) Marit snakker med.
 we know who C Marit talks with
 c. Vi vet hvem (*som) Pål tror (at) har kommet.
 we know who C Paul thinks that has come. (Taraldsen 1986)

In Defaka, the relevant head is always overt, but differently with subjects and non-subjects. Defaka has focus-movement. (38) does not involve it. (39a) involves local subject-focalization, (39b) object-

⁴See Bošković (2024) for more comprehensive accounts and additional arguments.

⁵For a different perspective on some such cases, see Pesetsky (2021).

focalization. Importantly, (39a-b) have different focus heads, with another difference in V-morphology: there is a morphological reflex of focalization in both cases, but it is different. Furthermore, long-distance-subject/adjunct-fronting patterns with object-fronting.

(38) *ì Bòmá ésé-kà-rè.*

I Boma see-fut-neg

‘I will not see Boma.’

(39) a. *ì kò Bòmá ésé-kà-rè.*

I foc Boma see-fut-neg

‘I will not see Boma.’

b. *Bòmá ndò ì ésé-kà-rè-kè.*

Boma foc I see-fut-neg-ke

‘I will not see Boma.’

(Bennett et al 2012)

There are also languages with local-subject-A’-extraction restricted processes. Dias (2022) shows Brazilian Portuguese *nunca*-ellipsis is possible under local-subject *wh*-movement, but not object/adjunct/long-distance-subject *wh*-movement.

(40) Marry kissed John.

Quem nunca [~~beijou~~ João]/*Quem Maria nunca [~~beijou~~ t]

who never kissed John who Mary never kissed

Like many sign languages, Hong Kong Sign language (HKSL) has rightward *wh*-movement: (41)-(42), involving *wh*-objects, show HKSL is not a *wh*-in-situ language. (SHORT-TIME means ‘quickly’.) Long-distance subject *wh*-movement patterns with objects (44)-(45). Local subject *wh*-movement is different (46): only locally-moved subjects needn’t undergo rightward *wh*-movement (Gan 2022 shows they do move).

(41) AARON EAT BREAKFAST/*WHAT SHORT-TIME.

(42) AARON EAT SHORT-TIME WHAT?

(43) *WHAT AARON EAT?

(44) KENNY THINK HELP AARON WHO?

‘Who does Kenny think helped Aaron?’

(45) *WHO KENNY THINK HELP AARON?

(46) WHO EAT FISH NOT?

(Gan 2002)

Lewis (2024) shows British English *do*-ellipsis allows only short subject *wh*-extraction, not object/long-distance-subject/adjunct-extraction, which is impossible to capture if they all land in the same position.

All this shows there are two *wh*-positions, with the lower *wh*-position occupied by locally-moved *wh*-subjects. Bošković (2024) argues this is a mixed A/A’-position on the border of A/A’-fields: it is the landing site of *wh*-movement (creating an Op-variable relationship), but also where the EPP is satisfied. This resolves the puzzle of *who left*, where the EPP seemed mysteriously voided. The EPP is satisfied: it is satisfied in the lower *wh*-position, a mixed A/A’-position confined to locally A’-moved subjects. (The A/A’-head has PF realization distinct from C in Norwegian and Defaka.)

Focalized subjects also move there. Consider (47).

(47) a. Only his girlfriend does John give any flowers.

b. *John gives only his girlfriend any flowers.

c. Only Mary showed any respect for the visitors.

(Branigan 1992:84)

The *only*-licensor c-commands the NPI in (47a-b), which cannot be licensed from a purely A-position, hence the ungrammaticality of (47b). (47c) is captured if the focalized subject also moves to the mixed

A/A'-position. *Only Mary* in (47c) is then not in Spec, IP, hence it can license the NPI, but it is also not in Spec, CP. Consequently, it allows inversion:

(48) Did only Mary show any respect for the visitors?

Bošković (2024) also argues for returning to early minimalist structure, which split IP into AgrsP/TP, addressing quirky subjects from this perspective. Based on subject-oriented-anaphor binding, I show quirky subjects are allowed only in *pro*-drop languages. The gist of the account is that the EPP can be satisfied in Spec, TP only in Chomsky's (2015) strong-T languages, which are *pro*-drop languages. Quirky subjects cannot go to Spec, AgrsP, which is reserved for agreeing nominative subjects, so they can satisfy the EPP only in *pro*-drop languages. All this leads to a conception of the EPP on a par with the contextual approach to phases in Bošković (2014), where there are phasal domains, and the highest phrase in a domain is a phase: there is an EPP domain (49), with the highest projection in this domain the locus of the EPP.

(49) [_{A/A'P} wh/A'-moved subjects [_{AgrsP} Amy [_{TP} quirky subjects

What is particularly important is that (49) captures mixed behavior of non-nominal subjects. Consider Locative Inversion (LI). It shows subject properties, like subject-raising (50) and the lack of WCO (51).

(50) On the wall_i seemed [_{t_i} to be hanging pictures].

(51) a. Into every flat_i peered its_i owner.
b. *Into every flat_i its_i owner peered.

LI also shows non-subject properties: it blocks extraction (52) and disallows inversion (53):

(52) ??*Who do you think that on this wall hung [pictures of t]?

(53) *Did on the wall hang pictures?

It also disallows inverse scope, like A'-moved subjects and in contrast to regular subjects:

(54) On some stage stood every actress. $\exists > \forall / * \forall > \exists$

All this follows if the LI moves to a higher subject position than regular subjects, i.e. Spec, A/A'P.⁶ It is debated whether LIs are subjects or topics due to conflicting data. The A/A'P-analysis resolves the conflict.

LI suggests non-nominal subjects move to Spec, A/A'P, not Spec, AgrsP/TP. Clausal subjects confirm this. They also show mixed subject properties, with a similar debate based on conflicting data as with LI (e.g. [*That Sue resigned*] seems likely vs **Is [that Sue resigned] likely*) The conflicting data are reconciled with the Spec, A/A'P-subject treatment.

LI and CP subjects indicate non-nominal subjects move to Spec, A/A'P. Interestingly, Hagstrom (1998) argues there is a QP above *wh*-DPs, which essentially makes them non-nominal. Below, I argue there is a similar projection above all phrases undergoing A'-movement, the right generalization being that non-DP/nominal subjects cannot move to Spec, AgrsP. The following then holds:⁷

(55) Non-nominal subjects satisfy the EPP in Spec, A/A'P.

(56) Agreeing nominative subjects satisfy the EPP in Spec, AgrsP.

(57) Non-agreeing/quirky nominal subjects satisfy the EPP in Spec, TP.

As noted above, there has been a move toward contextuality in many domains, including structure-building, labeling, the A/A'-distinction, locality (islandhood, phasehood, phasal edges), and spell-out. The EPP is

⁶See Bošković (2024) for why inversion is blocked with LI (but not with *only*-DPs).

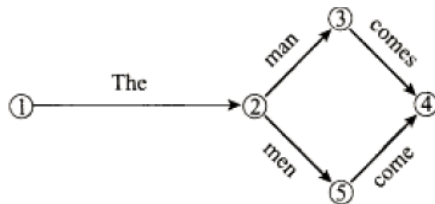
⁷Bošković (in press) argues that in languages where quirky subjects are subjects but less of a subject than in Icelandic, they are PPs located in Spec, A/A'P—the system thus captures “partial” quirky subjects.

now also added to this picture. In fact, the contextuality of the EPP is the same as with phases/phasal edges, being defined in the-highest-phrase-in-the-domain terms.

4.3. Markovian?

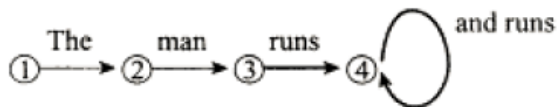
There is a larger point here: contextually-determined mechanisms cannot be considered simple Markovian properties. This confirms Chomsky's (1957) conclusion that finite-state machines (FSMs)/Markov processes are insufficient for language. FSMs are characterized by: initial/final state, specifications of transitions from one state to another, and symbols to print under transitions. (58) gives such a grammar, generating *the man comes* and *the men come*.

(58)



The transition from 1 to 2 prints *the*, the transition from 2 to 3 *man*, etc. The choice of a word restricts the next choice, so **The man come* is underivable. The hypothesis is that a speaker is an FSM. FSMs can capture infinity. Consider (59): there is a loop at 4, which yields *the man runs, and runs, and runs...*

(59)



Once it is in a state, the system knows that state, where it can go, and what symbol to print. It does not know where it was—it has no memory hence cannot capture dependencies that need memory. This is problematic. Thus, Lasnik et al (2000) note (60), which derives *missiles, anti-missile missiles, anti-anti-missile missile missiles...*, which cannot be captured by a Markovian device (MD).

(60) $\text{anti}^n \text{ missile}^{n+1}$

Long-distance dependencies (*the man who fired them is leaving*) are problematic for the same reason. MD thus cannot capture language, but does anything in language look like it? Yes: compositional semantics. When computing meaning we get by combining A and B, we don't know how we got to A or B—we only know A and B and what we get by combining them. In this respect, compositional semantics has no memory or look-ahead power—it's an MD. Many PF processes, like affix-hopping, require adjacency, which is Markovian by nature.

An MD is, however, insufficient for syntax. Probably the most important things the theory of syntax needs to capture are infinity and non-adjacent/long-distance relations. An MD captures the former, but not the latter. Returning to contextuality, contextuality is by its nature non-Markovian. Thus, to determine the phrase-structure status of like_2 in (3b) one needs to look at all of (3b); to determine whether α is a phasal edge in (1), one needs to know that there are no other XP-Specs and that XP is a phase, which is determined only when X merges with Y, with Y projecting. To determine the labeling status of α in (2) one needs to know *what* is a lower copy of a chain. None of this can be done in a Markovian fashion. The above discussion of contextuality thus confirms the non-Markovian nature of syntax, considerably broadening its scope.

Minimalism did endorse one Markovian property: no look-ahead. Under *local economy* (Collins 1996), when evaluating the best course of action, all we know is where we are and what we can do *next*,

with no look-ahead to the steps after the next step, which are Markovian properties. The contextual processes discussed above appear to require look-ahead but this is actually not the case. Bošković (2014) shows this for contextual phasehood and Bošković (2024) for contextual EPP. The gist of the latter is that Agrs, which has the EPP requirement, cannot attract non-nominal elements like PPs or *who* (which is not nominal). This triggers insertion of the A/A'-head, which can (and also has the EPP requirement). The A/A'-head and Agrs undergo feature-sharing (Frampton and Gutman 2002), so that when the EPP, treated as a feature, is satisfied with the A/A'-head, it is automatically satisfied on Agrs, with no look-ahead.

5. Adieu to the A/A'-distinction: Binding

The contextual EPP provides a natural mechanism for eliminating the A/A'-distinction, which is a GB remnant. Conceptually, it is as bad as the notion of government, which was abandoned long ago. In the spirit of Chomsky's (2021) duality of semantics, it would be natural if it were defined in terms of theta-roles, as a θ -non- θ -distinction. But this is not the case. Case is also relevant: A-positions are θ - and Case-positions (actually potential Case-positions). However, involving Case/ θ -roles is no longer enough to define A-positions. Spec, vP is assumed to be an A-position: it is the base-position of external arguments and the landing site of object shift. But Spec, vP through which *how* moves in (61) is an A'-position although Spec, vP is a potential Case/ θ -position.

(61) How_i do you [_{vP} t_i think [Mary left t_i]]?

Consider Japanese scrambling. (62) violates Condition A, with clause-mate scrambling in (63) taken to involve A-movement since Condition A is satisfied. Long-distance scrambling in (64) is assumed to involve A'-movement since Condition A is violated. However, long-distance scrambling in (65) is taken to involve A-movement, Condition A being satisfied.

(62) *[Otagai-no hahaoiya]-ga [Mary-to Pam]-ni atta.
each.other-GEN mother-NOM Mary-and Pam-DAT met
'Each other's mothers met Mary and Pam.'

(63) [Mary-to Pam]-ni_i [otagai-no hahaoiya]-ga t_i atta.

(64) *[Mary-to Pam]-ni_i [otagai-no hahaoiya]-ga [John-ga t_i atta to] omotteiru.
Mary-and Pam-DAT each.other-GEN mother-NOM John-NOM met C think
'Mary and Pam, each other's mothers think that John met.'

(65) John-to Bob_i-o otagai_i-no titioya_j-ga [_{CP} PRO_j t_i rikaisiyoo to] kokoromita.
John-and Bob-acc each.other's fathers-nom understand C attempted
'John and Bob, each other's fathers attempted to understand.'

It is simply not possible to relate the needed A/A'-distinction for (62)-(65) to Case/ θ -roles. There is no sense of Case/ θ -roles where the landing site of scrambling involves Case/ θ -positions in (63)/(65) but not (64). This is the tip of the iceberg, there are many other problematic paradigms. There is really no definition of the A/A'-distinction that works. Actually, there is: if movement creates a new binding configuration it is A-movement, if it does not it is A'-movement. This is the only "principled" way to make the cut. Its circularity is quite obvious.

The A/A'-distinction has been eliminated from relativized-minimality/intervention effects, which are now relativized with respect to the features involved. It would then be very appealing if it can be shown that binding effects can also be handled without it. This section attempts to do that. Its goal is the following: what we clearly need for binding is c-command, how about nothing else. This means that when a nominal is in a traditional A-position it can c-command, but when it's in a traditional A'-position the issue is the

lack of c-command: the fronted nominals do not c-command the anaphor in (66)-(67).⁸

(66) *Which students did each other's mothers meet?

(67) *John and Mary, each other's mothers met.

The basic idea is then rather straightforward: after undergoing "A'-movement", the nominal does not c-command the potential bindee hence cannot bind.

Regarding *wh*-phrases, it has been argued that there is a QP, projected by a Q-morpheme, above *wh*-DPs undergoing *wh*-movement (Hagstrom 1998, Bošković 2003, Cable 2010).⁹ The suggestion is then that there is a similar projection above all phrases undergoing traditional A'-movement (for topics, see Yoo 2018; for different but relevant discussion see Rezac 2003, Safir 2019). This is in fact why the relevant elements do not move to Spec, AgrsP (section 4)—they are not nominal. Importantly, the projections in question prevent the DP from c-commanding out—there is then no binding after traditional A'-movement.

The system discussed in section 4 provides evidence for the proposal that elements undergoing traditional A'-movement are not nominal. We have seen that non-nominal subjects satisfy the EPP in Spec, A/A'P, agreeing nominative subjects in Spec, AgrsP, and non-agreeing nominal subjects in Spec, TP. That traditional A'-subjects, like *who* in *who left*, pattern with non-nominal subjects in satisfying the EPP in Spec, A/A'P then provides evidence that they are indeed not nominal.

So, FocP, TopP, QP, not DP, is what undergoes traditional A'-movement. I take the relevant movements to be moving-element driven (see Bošković 2007)—if it is to be, for example, topicalized, TopP is projected right above it (what was a feature of the moving element driving movement in Bošković 2007 is now a projection), which then forces movement.

Lower down in the structure, the relevant elements do bind/c-command:

(68) John, Mary sold *t* pictures of himself.

(69) Who did Mary *t* sell pictures of himself?

Something then must happen during the derivation so that the relevant elements start as nominal but become non-nominal (see also Rezac 2003, Safir 2019). Something very similar in fact happens during the derivation of distributed coordination constructions (DCC) discussed in Bošković (2022). In (70), the individual conjuncts bear the external θ -roles but agreement is with the ConjP. Bošković (2022) (see also Zhang 2010) argues that the coordination in (70) is formed derivationally. The individual conjuncts are inserted in the respective θ -positions, then ConjP is formed through sideward merger, and inserted into the structure above the θ -positions. (71) confirms this. In (71a), ConjP binds the anaphor; in (71b) the individual conjuncts do. This indicates that when the anaphor is low, below where the relevant elements are generated, the conjuncts function as binders *separately*. When the anaphor is high, they bind *jointly*. (71c) shows the conjuncts must start separately—they induce a blocking effect separately (t_i and t_j are the closest subjects). The only way to account for (71) is to have the conjuncts start separately, with ConjP built during the derivation.

(70) [The dog_i and the rooster_j]_k were [_{VP} *t_i* barking] and [_{VP} *t_j* crowing] yesterday.

(71) a. [Bill and Sue]_i seem to each other_i to be the best candidate in the election and the best nominee for the convention respectively.

b. [Bill_j and Sue_k]_i hired himself_j and nominated herself_k respectively.

c. *[Bill_i and Sue_j]_k seem to be *t_i* the best candidate in each other's_k campaigns and *t_j* the best nominee in each other's_k parties respectively.

⁸Safir (2019) is an important predecessor of the current work in this respect. (It also uses derivational structure building in binding effects; for another way of eliminating the A/A'-distinction from binding, see van Urk 2015).

⁹I assume this is not the case with *wh*-phrases not undergoing *wh*-movement, see Bošković (2024).

This is in fact exactly what we need to achieve with derivational structure-building in our cases. What we need must then be independently possible, given the behavior of DCCs.

This, however, raises an issue regarding the cycle. Below I suggest several possibilities for implementing the required derivational structure building (X being the relevant element), which address this issue (for a different proposal see Safir 2019):

(a) We are dealing with sideward merger (with X), on a par with DCCs. Bošković (2022) argues that in DCC (70), after being introduced in their θ -positions, the conjuncts undergo sideward merger into ConjP, which is inserted back into the structure. On a par with this, the *wh*-DP is base-generated in the θ -position, and undergoes sideward-merger with X, with XP then reintroduced into the structure.¹⁰

(b) With DP already in the structure, X externally merges with it via adjunction, with X (an ambiguous head/phrase) projecting, which is a theoretical possibility. The DP ends up being adjoined to XP, but there is no cycle issue, since adjunction is not subject to it (Lebeaux 1988). Under Epstein's (1999) approach to c-command, where α c-commands what it merges with, the DP part does not c-command anything from the positions to which XP moves (only XP gains new c-command possibilities).

(c) Saito (2012) proposes heads can be base-generated in an adjunction structure. One of them then excorporates, undergoing head-movement (see also Martinović 2015, where a head is generated with features F1, F2... some of which can move out, projecting a phrase on their own). Thus, for Saito, *v* is base-adjoined to V and excorporates, projecting its own phrase after that. On a par with this, X is base-generated head-adjoined to D (as in Saito 2012) or it is a feature of D (as in Martinović 2015). X then excorporates, merging with DP, with X projecting; this is in a sense self-merger—it is possible that, as with adjunction, the cycle is not relevant in such cases. Alternatively, the cycle is phase-based—it can be “violated” within a phase (cf. Richards' 2001 tucking-in).

So, FocP/TopP/QP is projected above the DP during the derivation (blocking binding by the DP that was possible before their introduction), when the relevant elements need to undergo the movements in question, since this is what drives them. This separates semantic/information-structure-related functions: everything A-related is strictly formal and such formal licensing needs to be done before semantics/pragmatics-driven operations take place. This also provides a new take on Improper Movement (*A-A'-A): traditional A'-movement-driving features are not even present until all traditional A-movement is done, making Improper Movement trivially not possible.

The above captures in a natural way what it means to be a topic, for example. The relevant marking should be on the moving element, there is little sense in having it on a higher projection when that higher projection need not be there (when there is no topic), and when it is there, what should it attract? The closest what? To implement such attraction there needs to be a marking on the lower/moving element (X) indicating it is a topic (which might be all that's needed). During structure building, once purely formal properties are taken care of, if something will be a topic, it gets a topic projection and undergoes movement. When structure, where XP with a topic projection can be interpreted, is built, XP (i.e. TopP) merges with it.

Turning to additional cases, consider (72).

(72) Who seems to himself to be smart?

It has been argued that there is a phase with *seems* (Bošković 2015, Legate 2003), which requires SCM through its edge (cf. Legate's *Which of the papers he_i gave to Mary_j did every man_i__seem to her_j to dislike**). *Who* then stops at that phase edge as a DP in (72), binding *himself*, after which QP is added (cf.

¹⁰Bošković (2022) argues that the individual conjuncts undergo regular movement before sideward merger, which is not possible from θ -positions—this may also hold with XP, see (72).

fn 10 and the discussion of vP-internal scrambling below).¹¹

Notice the parallelism with DCCs, (73) being acceptable on a par with (72).¹²

(73) [Bill_i and Sue_j] seem to himself_i and to herself_j to be the best candidate in the election and the best nominee for the convention respectively.

Regarding (74), the problem is that *who*, which is still nominal at the edge of the *seems*-phase, is too close to the pronoun, causing a Condition B violation. This is confirmed by (75), where the distance is increased by embedding the pronoun.

(74) ?*Who_i seems to him_i to be smart?

(75) Who_i seems to his_i professor to be smart?

There is again parallelism with DCCs:

(76) ?*[Bill_i and Sue_j] seem to him_i and to her_j to be the best candidate in the election and the best nominee for the convention respectively.

(77) [Bill_i and Sue_j] seem to his_i supporters and to her_j donors to be the best candidate in the election and the best nominee for the convention respectively.

Some of the facts that motivated Ruys' (1994) observation that weak-crossover (WCO) violations improve by embedding the pronoun can also be captured with Condition B: there is a Condition B effect (with binding from the vP-phase edge) in (80a), but not (80b) due to pronoun embedding.

(78) a. *Which man_i did you give [a picture of t_i] to him_i?

b. Which man_i did you tell [a sister of t_i] that we hated him_i?

The A/A'-distinction is also not needed for (79)-(80): in (79), the anaphor's nominal part is embedded when the anaphor is higher than *Mary* (cf. fn 11)), so Condition C is not violated (Condition A is satisfied derivationally); in (80), *herself* remains a nominal, causing a Condition C violation.

(79) *Herself* Mary likes.

(80) **Herself* seems to Mary to be smart.

Regarding long-distance scrambling that does not create new binding configurations, it may be information-structure feature-driven, as in Miyagawa (1997)—no new binding configuration is then created due to the lack of c-command; see, however, Saito (2003), where due to the way copy-&-delete mechanism applies, only phonological features move into the higher clause in the relevant case—this analysis also does not require an A/A'-scrambling distinction, scrambling being a uniform operation.

Consider also WCO in (81): The nominal part of the *wh*-phrase does not c-command *his*, disallowing the bound-variable reading.¹³

(81) ?*Who_i does his_i mother like?

Strong crossover (**who_i does he_i like t_i*) can be handled as in the standard analysis.

Consider now Brazilian Portuguese (82). Islandhood/reconstruction indicate that without the case-marker *de*, the topic is base-generated; under the movement derivation *de* is needed.¹⁴ This is readily captured:

¹¹Spec, vP through which *which students* in (66) moves may need to be below subject θ -position (this is actually everyone's issue).

¹²(71a)/(73) indicate derivational ConjP formation occurs after movement to the *seems*-phase edge; ConjP can be merged into the structure in a higher Spec of this phase, or higher up in split IP.

¹³Regarding (51), (51a) instantiates the well-known fact that some PPs are porous for binding. In (51b), TopP above the PP blocks binding.

¹⁴(i)-(ii) (due to Renato Lacerda) illustrate the islandhood/reconstruction contrast.

under the movement derivation, the topic starts as a nominal—*de* is needed since nominals need Case. Under the base-generation derivation, the topic is never a nominal, which voids the Case requirement.

- (82) *Esse dinheiro/Desse dinheiro, eu vou precisar.*
 this money/of-this money I will need

Default Case is generally possible only in left-peripheral positions (if it were freely allowed it would void Case-filter effects, default Case being just a fancy term for unlicensed Case). The current system may explain this: only in left-periphery the relevant elements are not nominal.¹⁵

There are problematic cases where QR creates new variable binding possibilities: the quantifier c-commands the pronoun only after QR below.

- (83) Some daughter of [every author]_i hated some relative of his_i new wife. (Safir 2004)
 (84) That people hate him_i disturbs [every president]_i (Reinhart 1983)

This is problematic under traditional assumptions, which rely on the assumption that A'- but not A-movement induces WCO. QR should then induce WCO in (83)-(84). The problem does not arise in the current system, where all that matters is c-command: the quantifier c-commands the anaphor after QR. This means QR is not associated with introduction of an additional non-nominal projection, which is not surprising: the motivation for this projection is feature-checking. I have merely modified standard assumptions so that what was assumed to be a feature of YP undergoing movement is now a projection right above YP. QR is standardly assumed not to be feature-checking driven. There should then be no non-nominal projection above the quantifier, hence no c-command issue arises in (83)-(84).¹⁶

Lasnik and Stowell's (1991) weakest crossover is also captured. They note WCO unexpectedly does not arise in appositives, *tough*-movement, and parasitic gaps (85)-(87). It is standardly assumed that predication is crucially involved in these movements (for parasitic gaps, see also Nissenbaum 2000), which means they are motivated by semantic factors, like QR, not feature-checking driven. There is then no extra projection to block binding here. That the bolded elements in (85)-(87) bind from traditional A'-positions, a problem for standard assumptions where WCO arises with A'-movement, is now not an issue.

- (85) John_i, **who_i** his_i mother likes t_i,
 (86) Who/John_i will be easy for us [**Op_i** to get [his_i mother] to talk to e_i]
 (87) Who_i did you stay with t_i [**Op_i** before [his_i wife] had spoken to t_i]

Regarding anaphor binding, it may require a pre-spell-out phi-licensing Agree, which QR could not provide. The same may hold for the Op-movements in (86)-(87), given Sakamoto's (2020) arguments that null Op-movement is always covert. QR and null Op-movement then could not provide new anaphor binders. Interestingly, appositive movement can do that: (88a) is surprisingly better than (88b).

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- (i) *Esse/*desse livro, eu conheço [um aluno que tá precisando].*
 this/of-this book I know a student that is needing
 (ii) *Do/??o seu, melhor aluno, a Maria acredita que todo professor_i precisa em uma hora difícil.*
 of-the/the his best student the Mary believes that every professor needs in a hour difficult

¹⁵I assume **him_i, I like Bill_i* concerns competition effects, which are an independent issue (and not about c-command, see Jovović 2024). As for Condition C reconstruction (**which attack on Sue_i does she_i find unfair vs the attack on Sue_i seems to her_i unfair*), the A/A'-distinction here can be restated in nominal/non-nominal terms, or it can be captured with Chomsky's (1995) minimize-operator-restriction preference, relevant only to the "A'-example" (the issue is actually not simply about the A/A'-distinction).

¹⁶Quantifiers are quantifiers by their nature, not so with for example topics, which are "marked" as such with TopP. I leave open cases where bound-variable readings are disallowed (like Ruys' 2000 puzzling contrast *His_i mother gave [every soldier]_i a good luck charm to wear into battle vs ??His_i mother likes every boy_i*), merely noting that Ruys (2000) observes that "WCO intuitions are much stricter with *wh*-antecedents", which is unsurprising from the current perspective.

- (88) a. ?John and Mary, who each other's mothers like,
 b. *Which students did each other's mothers meet?

As discussed regarding (85), there is no additional projection with appositive movement; as a result, the movement both voids WCO and provides new anaphor binders.

Van Urk (2015) shows that what he calls topicalization in Dinka (the movement is associated with a variety of information-structural consequences, not just topicality) and analyzes as movement to Spec, CP can provide an anaphor binder even with long-distance movement. This can now be captured if XP is not projected with the movement in question, not surprisingly given the lack of a consistent discourse effect. (As van Urk notes, (91) argues against the A/A'-approach to binding.)

- (89) **Bòl**_i à-cíi [DP thùrá è **ròt-dè**_i] nyôoth [CP kè cùukù ___ t_iŋ].
 Bol 3s-prf.ov picture P self-sg.3sg show.nf C prf.1pl see.nf
 'Bol, a picture of himself has shown that we have seen.' (van Urk 2015:111)

Finally, in the current system, some A/A'-binding effects concern the timing of XP-insertion. In this respect, Masako Maeda (p.c.) observes that the system provides an account of the well-known but puzzling fact that vP-internal scrambling in Japanese (like DO-scrambling in [_{vP} DO_i IO t_i]) must count as A-movement (Saito 2003), if the above claim that XP can be inserted only after formal licensing is done is also interpreted to mean that XP cannot be inserted within the first phase where the relevant element is generated: vP-internal movement then cannot be A'-movement.

Concluding section 5, Bošković's (2024) approach to the EPP naturally opens the door for eliminating the A/A'-distinction from binding. In this system, Spec, A/A'P is a position reserved for non-nominal subjects. This is, however, where wh/focus/topic-subjects move, which provides evidence that wh/focus/topic-phrases are not nominal—this is why they cannot do traditional A-binding. Operators and argument phrases are thus distinguished categorially. The implementation was based on Bošković's (2007) idea that what triggers movement is the moved item, but what was a feature of the moving element driving movement in Bošković (2007) is now a projection above DP, preventing DP from binding/c-commanding out. An important part of the analysis was a parallelism with binding in DCCs, where binding possibilities change derivationally in a manner argued for elements undergoing traditional A'-movements.

6. Conclusion

The paper has highlighted the contextuality of syntax—there is a broader theoretical move toward contextuality that permeates many domains, like structure-building, labeling, islandhood, phases, phasal edges, spell-out domains, and the EPP. The contextuality of syntax was shown to be non-Markovian in nature, arguing against treating syntax as Markovian. One Markovian property was, however, maintained: no look-ahead.

Most importantly, we have seen that the contextual approach to the EPP opens the door for eliminating the A/A'-distinction from binding, in a way where all that is needed for binding is c-command: with traditional A'-movement, the issue is that the nominal simply fails to c-command the potential bindee.

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